



SYSTEM ANALYSIS & DESIGN

Management Information System Department
3rd Year - 2nd Semester

Lec #1

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SADAT ACADEMY FOR MANAGEMENT SCIENCES

About The Course

- Show mutual respect and listen.
- Everyone tried his best to attend the lecture on time.
- Please make your phone SILENT.
- Expect interaction.
- Avoid distractions.

- Course Name: *"System Analysis and Design"*.
- In this course, our materials will be based on the lectures presentations.
- Discussions:
 - [Whatsup](#) group.

■ Attendance & class participation	10
■ Assignments	20
■ Midterm	30
■ Final	40

- Define the terms system, system analysis, and system design.
- Describe the principal functions of the systems analyst.
- List and describe the phases of SDLC.
- Alternative approaches/methodologies of SLDC.
- Describe the various data gathering and analysis tools.
- Describe a selection of systems design tools.
- Explain the role of the maintenance task in SDLC.
- Increase students presentation and communication skills.
- Know how to work in teams.

General Speaking about SAD :(

- You plan to increase your sells.
- You want to reach large customers.
- You decide to make a website.
- From **thinking** in your idea to have your **actual** idea in the real-world, is system analysis and design.

Different Requirements Understanding



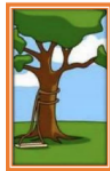
كيف وصفها العميل



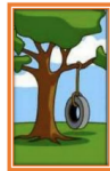
كيف فهمها مدير
المشروع



كيف صممها المحلل
القائم بالتحليل



كيف كتبها المبرمج



هذا ما أراده العميل في
الواقع

**Client : Why don't you clearly
understand my requirements ?**

His Requirement :



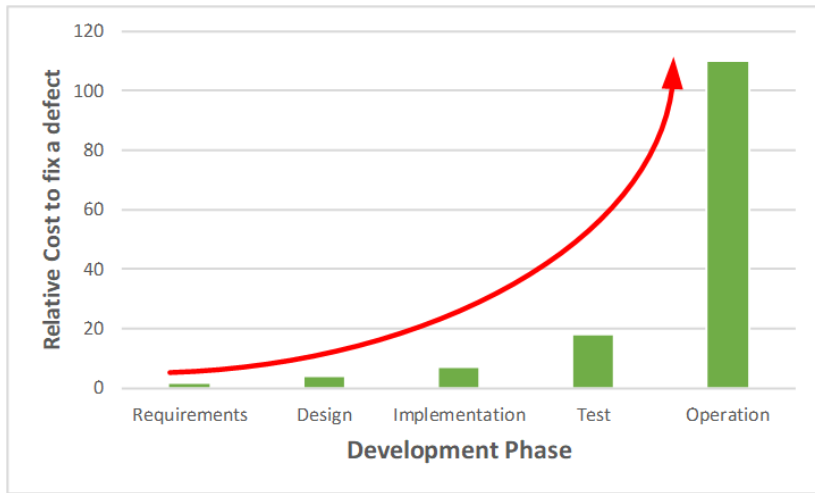
- It took the European Space Agency 10 years and 7 billion to produce Ariane 5, a giant rocket that intended to give Europe overwhelming supremacy in the commercial space business. The rocket was destroyed after 39 seconds of its launch, at an altitude of two and a half miles.



- Most errors (54%) are detected after coding and testing.
- Almost half of all errors in software (45%) are in requirements and design.
- Most errors (77%) made during requirements analysis are non-clerical.
- Need to do requirements and design right!

- A series of processes systematically undertaken to improve a business through the use of computerized information systems.
- Installing a system without proper planning leads to great user dissatisfaction and frequently causes the system to fall into disuse.
- Remember, finding and fixing a fault after software delivery is 100x more expensive than finding and fixing it during systems analysis or early design phases.

Failures Cost



- **System:**

A set of detailed methods, procedures and routines established or formulation to carry out specify activity, perform a duty or solve a problem.

- **System Analysis:**

The dissection of a system into its components to study how those components interact and work.

- **System Design:**

The process of defining the architecture, components, modules, interfaces and data for a system to satisfy specified requirements.

The System Analyst



Developer



Coder



Programmer



Software engineer

- An IT professional involved in the development of a computerized solution to a business problem.
- Works closely with all project team members so that the team develops the right system in an effective way.
- Serve as **change agent** who identify the organizational improvements needed, design systems to implement those changes, and train and motivate others to use the systems.

- Understanding what to change, knowing how to change it, and convincing others of the need for change require a wide range of skills.
- Skills categories:
 - 1 Technical,
 - 2 Business,
 - 3 Analytical,
 - 4 Interpersonal,
 - 5 Management, and
 - 6 Ethical.

- **Technical skills** to:

- Understand the organization's existing technical environment,
- The new system's technology foundation, and
- The way in which both can be fit into an integrated technical solution.

- **Business skills** to:

- Ensure that the IT delivers real business value.

- **Analytical skills to:**

- Parse data and information to develop creative, rational solutions.

- **Interpersonal skills to:**

- Communicate effectively, one-on-one with users and business managers and with programmers.
- Must be able to give presentations to large and small groups and to write reports.

- **Management skills** to:

- Manage people with whom they work, and
- Must manage the pressure and risks associated with unclear situations.

- **Ethical skills** to:

- Deal fairly, honestly, and ethically with other project team members, managers, and system users.
- Deal with confidential information that could cause harm (e.g., dissent among employees)

Task 1: Project Groups

- Teams up to 5:7 students.
- You are not allowed to change your team or team member once the teams have been formed.
- Project evaluation based on **Individual** contributions.
- Project selection will be based on the order of priority announced on the group.
- Use this link to preview some project ideas, or suggest your own idea.
<https://sites.google.com/view/dr-hamada-ibrhim/home>
- Complete the sending file "Assig#1 - Cover Page.doc"
 - 1 Title of your selected project.
 - 2 Team work (Name and email).
 - 3 Abstract about your project explaining project scope and intended work you will go on.